

Hybrid Piezoelectric and solar Powered AI-Enabled Smart Safety Shoe for Accident Detection and Emergency SOS

Mini Project Report

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By

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CERTIFICATE

This is to certify that the Mini project work entitled “**Hybrid Piezoelectric and solar Powered AI-Enabled Smart Safety Shoe for Accident Detection and Emergency SOS**” is being submitted by **N.VARSHASHRI(23AG1A7245)**, in partial fulfillment for the award of Degree of **BACHELOR OF TECHNOLOGY** in **ARTIFICIAL INTELLIGENCE & DATA SCIENCE (AI&DS)** to the Jawaharlal Nehru Technological University, Hyderabad during the academic year 2025-26 is a record of bonafide work carried out by them under our guidance and supervision.

The results embodied in this report have not been submitted by the student to any other University or Institution for the award of any degree or diploma.

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ABSTRACT

Exciting advancements in wearable technology have led to the creation of some innovative safety and tracking systems. However, many of the current options still rely on external charging and manual activation, which can be a bit tricky during emergencies. That's where our project comes in: we're introducing a Hybrid Piezoelectric and Solar Powered AI-Enabled Smart Safety Shoe, designed to be a self-powered and autonomous personal safety system.

This smart shoe harnesses energy from the pressure of walking using piezoelectric sensors, and it also includes a solar energy module for backup power, enhancing reliability over traditional single-source energy solutions. With an ESP32-based embedded IoT platform, it collects motion data through inertial sensors and uses AI to automatically identify falls and other accident scenarios.

When it senses an emergency, the system sends out SOS alerts along with real-time GPS location info via GSM communication, allowing for speedy rescue without needing a smartphone. By combining hybrid renewable energy sources, artificial intelligence, and IoT communication, our smart shoe offers a dependable solution for personal safety—perfect for accident prevention, enhancing women's safety, assisting the elderly, and handling outdoor emergencies.

The Smart Safety Shoe is an innovative wearable system designed to enhance personal safety by integrating advanced sensing, communication, and energy harvesting technologies. The primary objective of this project is to provide immediate assistance during emergency situations such as accidents or sudden falls, especially for women, elderly individuals, and people in isolated environments.

The system is built around the ESP32 microcontroller, which acts as the central processing unit and coordinates all operations. It continuously monitors the user's movement using the MPU6050 sensor, which detects abnormal activities such as sudden falls. In case of an emergency, the system automatically triggers an alert. Additionally, a manual SOS button is provided to allow the user to send an alert instantly.

For real-time communication, the system uses a GPS module to obtain the user's location and a GSM module to send alert messages to predefined contacts. This ensures quick response and timely assistance during critical situations. One of the key features of this project is the integration of a hybrid power system. The piezoelectric sensor generates electrical energy from walking pressure, while a solar panel provides additional power, ensuring continuous and energy-efficient operation.

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CHAPTER-1

INTRODUCTION

1.1 PROJECT OVERVIEW

The Smart Safety Shoe with Emergency Alert System and Piezoelectric Energy Generation is an innovative wearable solution designed to enhance personal safety and demonstrate energy harvesting. This project integrates an ESP32 microcontroller with multiple modules including MPU6050 for motion sensing, GPS for real-time location tracking, GSM for sending emergency alerts, and piezoelectric sensors embedded within the shoe sole.

The system is capable of detecting emergency situations through both manual and automatic methods. A manual SOS button allows the user to trigger an alert instantly, while the MPU6050 sensor enables automatic detection of abnormal motion or falls. In such cases, the system sends an emergency message along with the user's live location to predefined contacts using the GSM module.

In addition to safety features, the project also demonstrates energy generation using piezoelectric sensors. When pressure is applied during walking or stepping, these sensors generate small electrical energy, which is monitored using the ESP32 and displayed as varying values. This showcases the concept of converting mechanical energy into electrical energy.

The entire system is powered by a rechargeable Li-ion battery, which is charged using a solar panel through a TP4056 charging module, ensuring a sustainable and continuous power supply.

For communication, the system uses a GPS module to obtain real-time location data and a GSM module to send alert messages to predefined contacts. This enables quick tracking and faster response during emergencies. One of the unique features of this project is the integration of energy harvesting techniques. Piezoelectric sensors generate electrical energy from walking pressure, while a solar panel provides additional power, ensuring continuous operation and improved battery efficiency.

The entire system is designed to be compact, lightweight, and wearable, making it suitable for everyday use. It is especially useful for women, elderly individuals, and people working in isolated environments. By combining safety monitoring, real-time communication, and sustainable energy solutions, the project offers a practical and cost-effective approach to personal security. The Smart Safety Shoe is an innovative wearable system designed to enhance personal safety and ensure real-time emergency response. The project integrates multiple technologies such as embedded systems, sensors, communication modules, and energy harvesting mechanisms into a single compact device. The main objective of the system is to provide immediate assistance during emergency situations like falls, accidents, or unsafe conditions.

1.2 OBJECTIVES

- To design and develop a smart safety shoe that enhances user protection in emergency situations.
- To implement both manual and automatic SOS alert systems using a push button and motion sensing (MPU6050).
- To enable real-time location tracking using a GPS module for accurate emergency response.
- To send emergency alerts via GSM module to predefined contacts with location details.
- To integrate piezoelectric sensors for generating electrical energy from human motion (walking/pressure).
- To demonstrate the concept of energy harvesting by converting mechanical energy into electrical energy.
- To ensure continuous operation using a rechargeable Li-ion battery powered by a solar charging system.
- To build a compact, wearable, and efficient system suitable for real-world safety applications.
- To develop a cost-effective and reliable solution compared to existing systems.
- To enable real-time location tracking using a GPS module for accurate position identification.
- To ensure the system is low power consuming and energy-efficient for long-term usage.
- To improve safety and security for women, elderly people, and individuals in isolated environments.
- To implement a hybrid power system using both piezoelectric and solar energy for continuous operation.
- To utilize renewable energy sources for reducing dependency on external charging.
- To improve battery life and power management using hybrid energy solutions.
- To design a system that is user-friendly and easy to operate in real-time situations.
- To create a solution that can be expanded or upgraded in future with additional features like IoT or mobile app integration.
- To ensure fast and reliable communication between the user and emergency contacts.
- To design a system capable of detecting emergency situations automatically without human intervention.
- To provide location-based emergency alerts for quick rescue operations.

1.3 PROBLEM FORMULATION

In today's fast-paced world, ensuring personal safety, especially during emergencies such as accidents or sudden falls, is a major concern. Many individuals, including elderly people and those working in isolated environments, may not be able to seek immediate help during critical situations. Existing safety systems often rely on manual intervention, lack real-time location tracking, or are not integrated into everyday wearable devices.

Additionally, with increasing demand for sustainable solutions, there is a need to explore alternative energy sources that can utilize everyday human activities, such as walking, to generate electrical energy. However, most current systems do not effectively combine safety features with energy harvesting in a compact and practical manner.

Therefore, there is a need to develop a smart, wearable system that can automatically detect emergencies, send real-time alerts with location, and also demonstrate energy generation using human motion, all within a single integrated solution.

In recent years, personal safety has become a major concern, especially for elderly individuals, women, and people working in isolated environments. During emergency situations such as accidents, falls, or sudden health issues, it is often difficult for individuals to seek immediate help. Traditional safety systems mainly depend on manual intervention, where the user must actively trigger an alert. However, in many real-life situations, the user may be unconscious or unable to respond, making these systems ineffective.

Although some modern systems use sensors for fall detection, they often lack reliable communication features such as real-time location tracking and instant alert messaging. Without accurate location information, providing timely assistance becomes difficult. Additionally, many existing solutions are not designed as wearable systems, making them inconvenient for continuous use in daily life.

Another major issue is related to power consumption and sustainability. Most systems rely entirely on battery power, which requires frequent charging and limits long-term usage. While energy harvesting techniques such as piezoelectric generation have been explored, they alone are not sufficient to power the system continuously. Also, the lack of hybrid energy solutions reduces system efficiency and reliability.

Furthermore, there is a lack of integration among safety features, communication modules, and energy harvesting systems. Existing solutions tend to focus on only one aspect, rather than providing a complete and practical solution. This creates a need for a system that is compact, wearable, energy-efficient, and capable of automatic emergency detection with real-time communication.

1.4 EXISTING SYSYTEM

The existing systems in the field of wearable safety and energy harvesting primarily focus on individual functionalities rather than an integrated approach. Many safety systems rely on manual emergency alert mechanisms, where the user must press a button to send an alert, which may not be effective in situations where the user is unconscious or unable to respond. Some systems use motion sensors for fall detection, but they often lack real-time communication features such as GPS tracking and GSM-based alert messaging.

On the other hand, energy harvesting systems using piezoelectric materials have been developed to generate electrical energy from human motion. These systems demonstrate the concept of converting mechanical energy into electrical energy; however, the generated power is usually insufficient for continuous operation and is often not integrated with practical applications like safety monitoring. Additionally, most existing systems do not utilize hybrid energy solutions such as combining solar and piezoelectric energy, leading to limited efficiency.

Furthermore, many current solutions are bulky, power-consuming, or not designed for wearable convenience, making them less practical for everyday use. Due to these limitations, existing systems fail to provide a complete solution that combines automatic emergency detection, real-time communication, and sustainable energy generation in a compact and efficient manner.

- ❖ Current safety solution mainly include moblie emergency appliactions, GPS tracking wearables, and smart watches with SOS features.
- ❖ These systems provide location tracking and manual emergency alert functionality for users.
- ❖ Fitness smart shoes and wearable devices focus primarily on activity monitoring, health tracking, or navigation support rather than accident response.
- ❖ Existing energy harvesting footwear research mainly concentrate on power generation using piezoelectric materials without integrating intelligent safety mechanisms.
- ❖ most available systems rely heavily on manual activation, external charging, and smartphone connectivity for operation.

1.5 PROPOSED SYSTEM

The proposed system is a Smart Safety Shoe integrated with an Emergency Alert System and Energy Harvesting mechanism, designed to enhance personal safety and provide a sustainable power solution. The system uses an ESP32 microcontroller as the central unit to coordinate all operations including sensing, processing, and communication.

The system incorporates both manual and automatic emergency detection mechanisms. A push button is provided for manual SOS activation, allowing the user to send an alert instantly in critical situations. In addition, the MPU6050 sensor continuously monitors motion and detects abnormal activities such as sudden falls or lack of movement. When such conditions are identified, the system automatically triggers an emergency alert.

For communication, the system integrates a GPS module to obtain real-time location and a GSM module to send SMS alerts to predefined contacts. This ensures that the user's exact location is shared quickly during emergencies, enabling faster assistance.

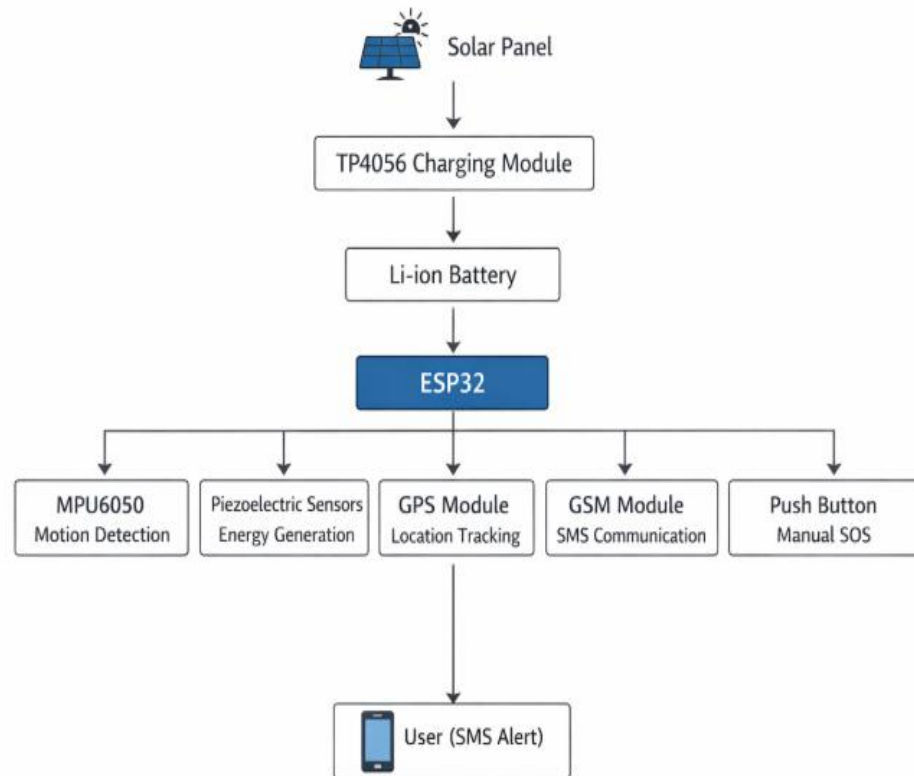
A key feature of the proposed system is the integration of piezoelectric sensors, which generate electrical energy from pressure applied during walking. This demonstrates energy harvesting by converting mechanical energy into electrical energy. However, since piezoelectric energy alone is limited, the system also includes a solar panel-based charging system. The solar panel charges a Li-ion battery through a TP4056 module, ensuring continuous and reliable power supply.

The overall system is designed to be compact, wearable, energy-efficient, and reliable, making it suitable for real-world applications such as elderly care, women safety, and personal security.

Additional features:

- ❖ Development of a Hybrid Piezoelectric and Solar Powered AI-Enable Smart Safety Shoe ensuring continuous operation through renewable energy harvesting.
- ❖ Piezoelectric sensors generate electrical energy from walking pressure, while a solar panel provides stable supplementary charging.
- ❖ AI-based motion analysis using accelerometer and gyroscope data distinguishes normal movement from falls, accident impacts, and abnormal inactivity conditions.
- ❖ Implementation of a Multi-Trigger SOS Emergency System including manual SOS activation, automatic fall detection, accident detection, and no-movement confirmation alerts.
- ❖ Real-time GPS location tracking integrated with GSM communication to automatically send emergency alerts to predefined contacts.
- ❖ Designed as a self-powered, autonomous wearable safety system operating independently without smartphone dependency.

1.6 BLOCK DIAGRAM



1.7 LIST OF COMPONENTS

Core Controller	Power Management
➤ ESP32 DevKit V1 (ESP-WROOM-32 Microcontroller)	➤ 18650 Li-ion Rechargeable Battery (3.7V, 2600mAh) ➤ TP4056 Battery Charging Module (Type-C Protection Version) ➤ Bridge Rectifier (DB107 / 1N4007 Diodes) ➤ Electrolytic Capacitor (1000μF)
Energy Harvesting Components	Control & Interface
➤ Piezoelectric Disc Sensors (27mm Ceramic Type, Qty: 10–15) ➤ Flexible Solar Panel (5V, 2W Monocrystalline)	➤ Push Button Switch (Manual SOS Trigger) ➤ Connecting Wires, Breadboard / PCB Assembly
Sensors & Communication Modules	
➤ MPU6050 Accelerometer & Gyroscope Sensor (AI Motion Detection) ➤ NEO-6M GPS Module (Location Tracking) ➤ SIM800L GSM Module(SOS Communication)	

CHAPTER-2

LITERATURE SURVEY

With the rapid growth of wearable technology and embedded systems, there has been a significant increase in research focused on developing smart devices that enhance personal safety and energy efficiency. Wearable safety systems are especially important for women, elderly individuals, and people working in isolated environments, where immediate assistance may not always be available. In recent years, researchers have explored various approaches involving sensors, communication technologies, and energy harvesting techniques to address these challenges. This section presents a detailed review of existing works related to smart shoes, safety systems, and energy harvesting technologies.

The study titled “Development of Smart Shoes Using Piezoelectric Material” (2020) focused on designing a smart shoe capable of generating electrical energy through piezoelectric materials. The research demonstrated that mechanical pressure generated during activities such as walking, jogging, and jumping can be effectively converted into electrical energy. In this system, piezoelectric sensors were embedded inside the shoe sole, and the generated energy was stored using capacitors to power low-energy electronic devices. The study also compared different piezoelectric materials such as ceramic and polymer, concluding that higher energy output was achieved during high-intensity activities like jumping. Although the system successfully demonstrated energy harvesting, it mainly concentrated on power generation and did not include safety features such as emergency communication, SOS alerts, or intelligent decision-making. Furthermore, the generated energy was not sufficient for continuous real-world applications, and the system lacked hybrid energy integration such as solar power [1].

Another study titled “Smart Shoe for Elderly Tracking and Rescue with Piezoelectric Based Energy Harvesting System” (2023) proposed a wearable system aimed at improving elderly safety. This system utilized piezoelectric plates to generate energy from foot movement and incorporated GPS for real-time location tracking along with GSM communication to send periodic location updates via SMS. The generated energy was used to power low-energy modules, enhancing mobility and independence for elderly users. While the system improved tracking capabilities, it primarily focused on location monitoring rather than emergency detection. It lacked automatic fall detection and did not include an SOS triggering mechanism during critical situations. Additionally, the system did not implement intelligent decision-making techniques and relied heavily on piezoelectric energy without integrating hybrid energy sources such as solar power. As a result, its efficiency and reliability in real-time emergency scenarios were limited [2].

In addition to these studies, several research works have explored wearable safety systems using sensors such as accelerometers and gyroscopes to detect abnormal movements and accidents. These systems are capable of identifying sudden changes in motion patterns and have shown promising results in fall detection applications. However, many of these systems suffer from limitations such as inaccurate detection due to improper calibration, environmental disturbances, or noise in sensor data. Moreover, several systems rely on manual activation of alerts, which reduces their effectiveness in situations where the user is unconscious or unable to respond [3].

On the other hand, piezoelectric-based energy harvesting systems have been widely researched for their ability to convert mechanical energy from human motion into electrical energy. These systems demonstrate the feasibility of generating power from daily activities such as walking and running. However, the amount of energy generated is generally low and insufficient to support the continuous operation of electronic systems. Therefore, piezoelectric energy is often used as a supplementary power source rather than a primary energy solution. Researchers have attempted to improve efficiency, but standalone piezoelectric systems still face practical limitations in real-world applications [4].

Furthermore, modern safety systems often incorporate GPS and GSM technologies to enable real-time tracking and communication. These systems improve user safety by allowing location-based alert transmission during emergencies. However, many of these systems lack automation and depend on manual triggering of alerts. This reduces their reliability in critical situations. Additionally, most of these systems do not implement hybrid energy solutions, such as combining solar energy with piezoelectric generation, which could significantly enhance system efficiency and sustainability [5].

Recent advancements also highlight the integration of solar energy in wearable systems to provide a more reliable and consistent power source. Solar panels can generate higher energy compared to piezoelectric sensors, but their performance depends on environmental conditions such as sunlight availability. Very few research works have successfully combined both solar and piezoelectric energy to create an efficient hybrid system for continuous operation.

Overall, the literature clearly indicates that existing systems focus on individual aspects such as safety monitoring, communication, or energy harvesting, but lack complete integration of all these features into a single system. There is a significant research gap in developing a compact, wearable solution that combines automatic emergency detection, real-time communication, and efficient hybrid energy management.

Therefore, the proposed Smart Safety Shoe system addresses these limitations by integrating sensor-based safety mechanisms with both piezoelectric and solar energy sources. This approach provides a more efficient, reliable, and practical solution for real-world applications, ensuring enhanced personal safety along with sustainable power utilization.

CHAPTER-3

INTRODUCTION TO HARDWARE

3.1 WORKING PRINCIPLE

The working of the Smart Safety Shoe system is based on sensing, processing, and communication. The ESP32 continuously monitors input from sensors such as the MPU6050 and piezoelectric sensors. When a fall or abnormal motion is detected, or when the user presses the SOS button, the system processes this input and triggers an emergency alert.

The GPS module retrieves the real-time location of the user, and the GSM module sends this information via SMS to predefined contacts. Simultaneously, the piezoelectric sensors embedded in the shoe generate electrical energy when pressure is applied during walking, demonstrating energy harvesting. The solar panel ensures continuous power supply by charging the battery.

3.1.1 Functional Requirements

- The functional requirements define what the system should do:
- The system must detect emergency situations using motion sensing (MPU6050).
- The system must allow manual SOS activation through a push button.
- The system must acquire real-time location using a GPS module.
- The system must send alert messages using a GSM module.
- The system must generate electrical energy using piezoelectric sensors when pressure is applied.
- The system must monitor and process sensor data using the ESP32 microcontroller.
- The system must operate using a rechargeable battery with solar charging support.

3.1.2 Non-Functional Requirements

- The non-functional requirements define system quality and performance:
- The system should be compact and lightweight for comfortable wearable use.
- The system should be reliable and accurate in detecting emergencies.
- The system should have low power consumption for longer operation.
- The system should provide quick response time for sending alerts.
- The system should be durable to withstand daily usage in a shoe environment.
- The system should be easy to use and maintain.
- The system should ensure continuous operation through solar-assisted charging.

3.2 HARDWARE COMPONENTS DESCRIPTION

3.2.1 ESP32 Microcontroller

3.2.1.1 Introduction

The ESP32 is a powerful and low-cost microcontroller with built-in Wi-Fi and Bluetooth capabilities. It acts as the central processing unit of the Smart Safety Shoe system, responsible for collecting data from sensors, processing it, and controlling communication modules such as GPS and GSM. Due to its high performance and low power consumption, ESP32 is widely used in IoT-based applications.

3.2.1.2 Features

- Dual-core processor for faster performance
- Built-in Wi-Fi and Bluetooth
- Multiple GPIO pins for sensor interfacing
- Analog-to-Digital Converter (ADC) support
- Low power consumption
- High processing capability



3.2.1.3 Pin Configuration

The ESP32 consists of multiple GPIO pins used for connecting sensors and modules:

- GPIO34 – Analog input (used for Piezo sensor)
- GPIO21, GPIO22 – I2C communication (MPU6050)
- TX/RX pins – Used for GSM and GPS communication
- 3.3V & GND – Power supply

3.2.1.4 Working

The ESP32 continuously reads data from sensors such as MPU6050 and piezoelectric sensors. Based on the input conditions, it processes the data and triggers actions like sending alerts through the GSM module. It also monitors energy generated by the piezo sensors and controls overall system functionality.

3.2.2 MPU6050 SENSOR

3.2.2.1 Introduction

The MPU6050 is a motion tracking sensor that combines a 3-axis accelerometer and a 3-axis gyroscope in a single module. It is used in the Smart Safety Shoe system to detect motion, orientation, and sudden changes such as falls. This sensor plays a crucial role in enabling automatic emergency detection.

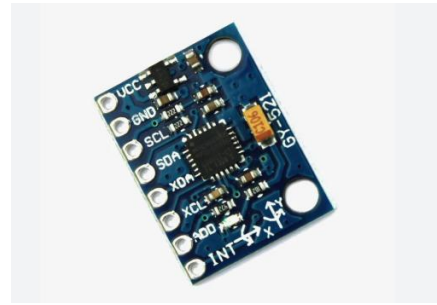
3.2.2.2 Features

- 3-axis accelerometer and 3-axis gyroscope
- Digital output using I2C communication
- High accuracy motion sensing
- Low power consumption

3.2.2.3 Pin Configuration

The MPU6050 module has the following pins:

- VCC – Power supply (3.3V)
- GND – Ground
- SDA – Data line (connected to ESP32 GPIO21)
- SCL – Clock line (connected to ESP32 GPIO22)



3.2.2.4 Working

The MPU6050 continuously measures acceleration and angular velocity along three axes. The accelerometer detects linear motion, while the gyroscope detects rotational movement. The ESP32 reads this data through I2C communication and analyzes it to identify abnormal conditions such as sudden falls or unusual movement patterns. Based on this data, the system can trigger emergency alerts automatically.

3.2.2.5 Applications

- Fall detection systems
- Motion tracking devices
- Wearable electronics
- Robotics and gaming systems

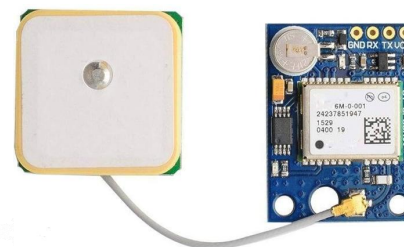
3.2.3 GPS MODULE

3.2.3.1 Introduction

The GPS (Global Positioning System) module is used to determine the real-time location of the user. In the Smart Safety Shoe system, the GPS module plays a crucial role in tracking the user's position during emergency situations. It provides accurate latitude and longitude coordinates, which are sent to predefined contacts through the GSM module.

3.2.3.2 Features

- Provides real-time location data
- High accuracy positioning
- Supports satellite-based navigation
- Compact and low power consumption
- Easy interfacing with microcontrollers



3.2.3.3 Pin Configuration

The GPS module consists of the following pins:

- VCC – Power supply
- GND – Ground
- TX – Transmits data to ESP32
- RX – Receives data from ESP32

3.2.3.4 Working

The GPS module communicates with satellites to obtain location data such as latitude and longitude. This data is sent to the ESP32 through serial communication. The ESP32 processes this information and converts it into a readable format or a Google Maps link. During an emergency, this location data is included in the alert message sent to the user's contacts.

3.2.3.5 Applications

- Vehicle tracking systems
- Navigation systems
- Emergency response systems
- Location-based services

3.2.4 GSM MODULE

3.2.4.1 Introduction

The GSM (Global System for Mobile Communication) module is used for sending alert messages in emergency situations. In the Smart Safety Shoe system, the GSM module enables communication by sending SMS notifications along with location details to predefined contacts. It ensures that help can be reached quickly during critical conditions.

3.2.4.2 Features

- Supports SMS communication
- Wide network coverage
- Reliable and fast message transmission
- Easy interfacing with microcontrollers
- Operates on standard SIM card

3.2.4.3 Pin Configuration

The GSM module consists of the following pins:

- VCC – Power supply
- GND – Ground
- TX – Transmits data to ESP32
- RX – Receives data from ESP32



3.2.4.4 Working

The GSM module communicates with the ESP32 using serial communication. When an emergency is detected (manual or automatic SOS), the ESP32 sends AT commands to the GSM module. The GSM module then sends an SMS containing the alert message along with the GPS location link to predefined phone numbers.

3.2.4.5 Applications

- Emergency alert systems
- SMS-based communication systems
- Remote monitoring systems
- Security systems

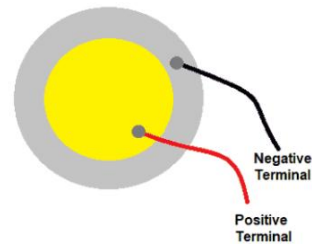
3.2.5 PIEZOELECTRIC SENSOR

3.2.5.1 Introduction

The piezoelectric sensor is used to generate electrical energy from mechanical pressure. In the Smart Safety Shoe system, piezo sensors are embedded inside the shoe sole to convert the pressure produced during walking into electrical signals. This helps in demonstrating energy harvesting from human motion.

3.2.5.2 Features

- Converts mechanical pressure into electrical energy
- Small and lightweight
- No external power required
- Fast response to pressure changes
- Durable and suitable for wearable applications



3.2.5.3 Pin Configuration

The piezo sensor has two terminals:

- Positive terminal → Connected to ESP32 analog pin (GPIO34)
- Negative terminal → Connected to GND

3.2.5.4 Working

When pressure is applied on the piezo sensor, it generates a small electrical voltage due to the piezoelectric effect. This voltage is read by the ESP32 through an analog pin. The generated values vary depending on the force applied, where higher pressure produces higher voltage values. This demonstrates the concept of converting mechanical energy into electrical energy.

3.2.5.5 Applications

- Energy harvesting systems
- Pressure sensing devices
- Smart wearable devices
- Footstep energy generation

3.2.6 SOLAR PANEL AND POWER SYSTEM

3.2.6.1 Introduction

The solar panel and power system are responsible for supplying energy to the Smart Safety Shoe system. A rechargeable Li-ion battery is used as the main power source, which is charged using a solar panel through a TP4056 charging module. This setup ensures continuous and sustainable operation of the system.

3.2.6.2 Features

- Renewable energy source using solar power
- Rechargeable Li-ion battery
- Efficient charging using TP4056 module
- Provides continuous power supply
- Supports portable and wearable design

3.2.6.3 Components in Power System

- Solar Panel
- TP4056 Charging Module
- Li-ion Battery

3.2.6.4 Working

The solar panel converts sunlight into electrical energy. This energy is fed into the TP4056 charging module, which regulates the charging process and safely charges the Li-ion battery. The battery then supplies power to all components of the system, including the ESP32, sensors, GPS, and GSM modules. This ensures uninterrupted operation even when external power sources are not available.

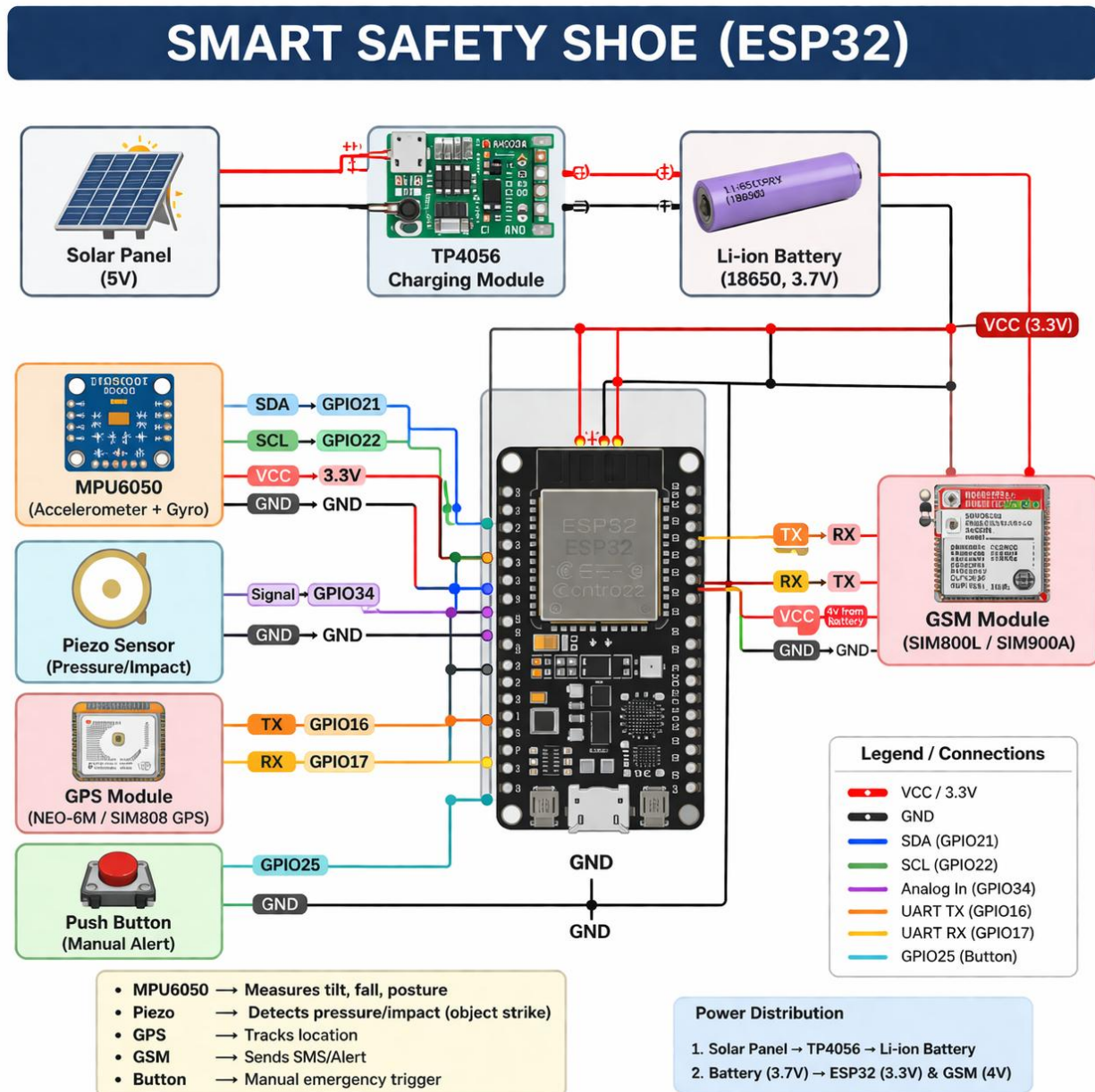
3.2.6.5 Applications

- Solar-powered wearable devices
- Portable electronic systems
- Energy-efficient IoT devices
- Sustainable power solutions



3.3 SCHEMATIC DIAGRAM

The schematic diagram represents the overall electrical connections of the Smart Safety Shoe system. It shows how the ESP32 microcontroller is connected to different components such as the MPU6050 sensor, GPS module, GSM module, piezoelectric sensor, and push button. The MPU6050 is connected through I2C communication, while GPS and GSM modules are connected using serial communication. The piezo sensor is connected to an analog pin to measure the voltage generated from pressure. The system is powered by a Li-ion battery, which is charged using a solar panel through a TP4056 charging module. This diagram helps in understanding how all components are interconnected and how the system functions as a whole.



CHAPTER-4

SOFTWARE DESIGN

4.1 UML DIAGRAMS

UML stands for Unified Modelling Language. UML is a standardized general purpose modelling language in the field of object-oriented software engineering. The Standard is managed, and was created by, the Object Management Group. The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form UML is comprised of two major components: a Meta- model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

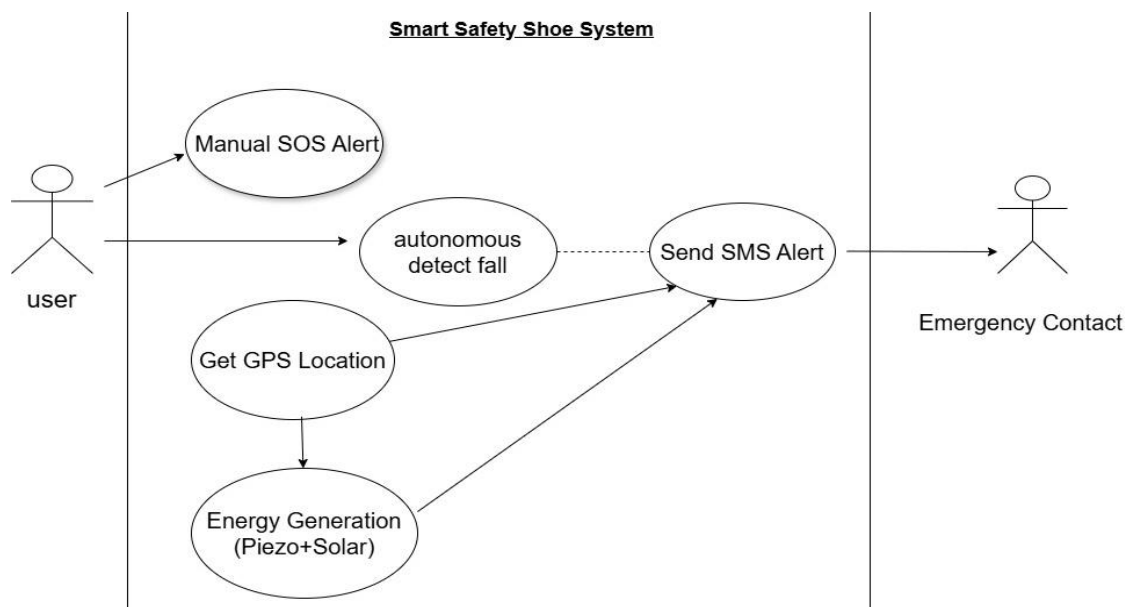
In this project, different types of UML diagrams such as Use Case Diagram, Class Diagram, Sequence Diagram, and Activity Diagram are used to describe various aspects of the system.

4.1.1 Usecase Diagram

A Use Case Diagram is a behavioral UML diagram that represents the interaction between the user (actor) and the system. It helps in identifying the functional requirements of the system by showing different operations that the system can perform. This diagram provides a clear understanding of how the user interacts with the Smart Safety Shoe system.

In this project, the primary actor is the user who interacts with the system during normal and emergency situations. The main use cases include detecting a fall, sending an emergency alert, tracking the user's location, and generating energy through piezoelectric and solar sources.

When a fall is detected automatically by the sensor or when the user presses the SOS button manually, the system processes the input and triggers an alert. The GPS module retrieves the location, and the GSM module sends an emergency message to predefined contacts.

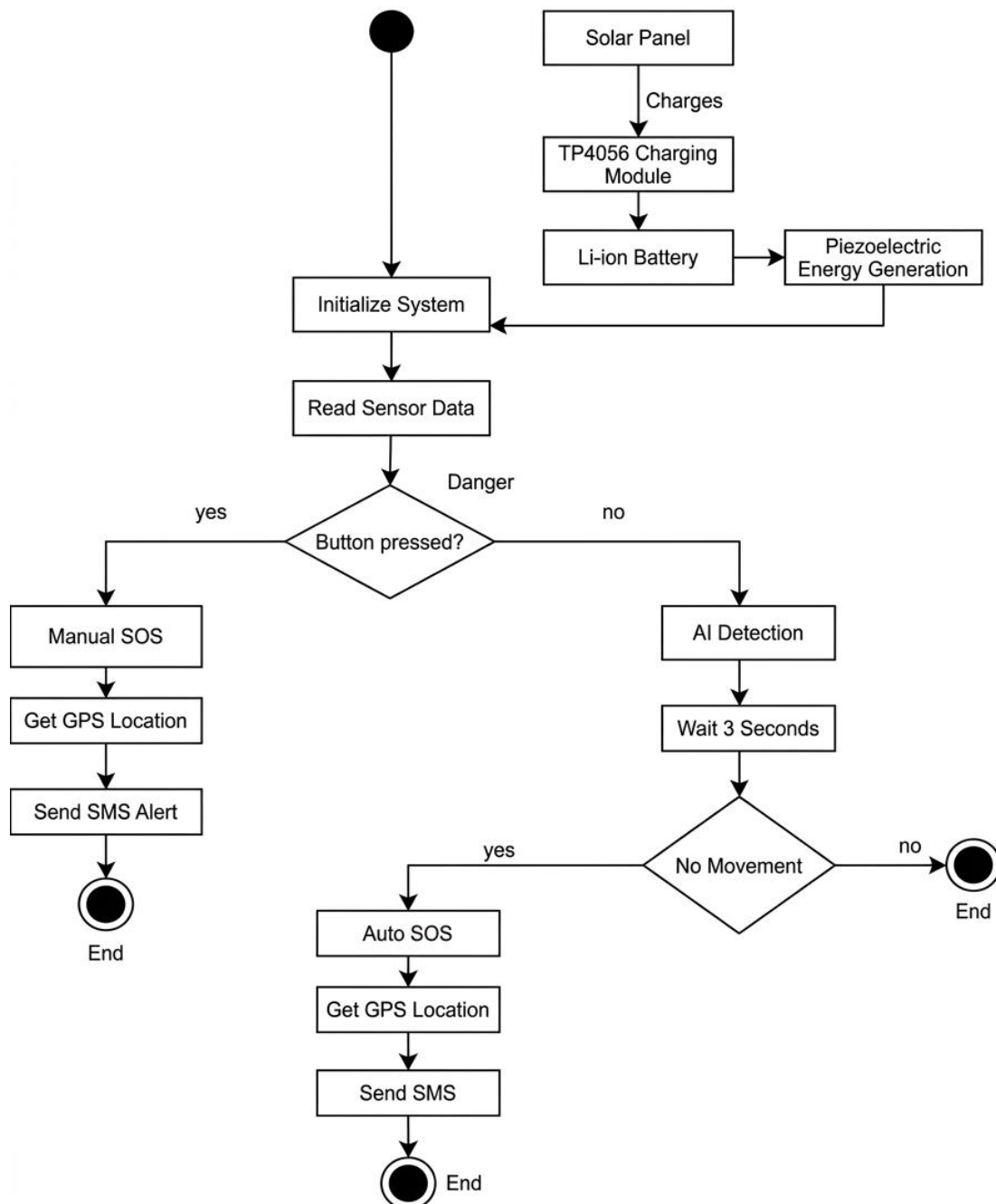


4.1.2 Activity Diagram

In the Smart Safety Shoe system, the activity diagram begins with the system initialization, followed by continuous monitoring of sensor data. The MPU6050 sensor checks for any abnormal motion or fall detection. If a fall is detected, the system proceeds to retrieve the user's location using the GPS module.

Then, the GSM module sends an emergency alert message to predefined contacts. If no abnormal condition is detected, the system continues monitoring the user's activity.

The diagram also includes a manual path where the user can press the SOS button to trigger an alert directly. Decision nodes are used to represent conditions such as "Fall detected?" or "Button pressed?", making the workflow clear and easy to understand. Overall, the Activity Diagram provides a clear representation of the system's working process,



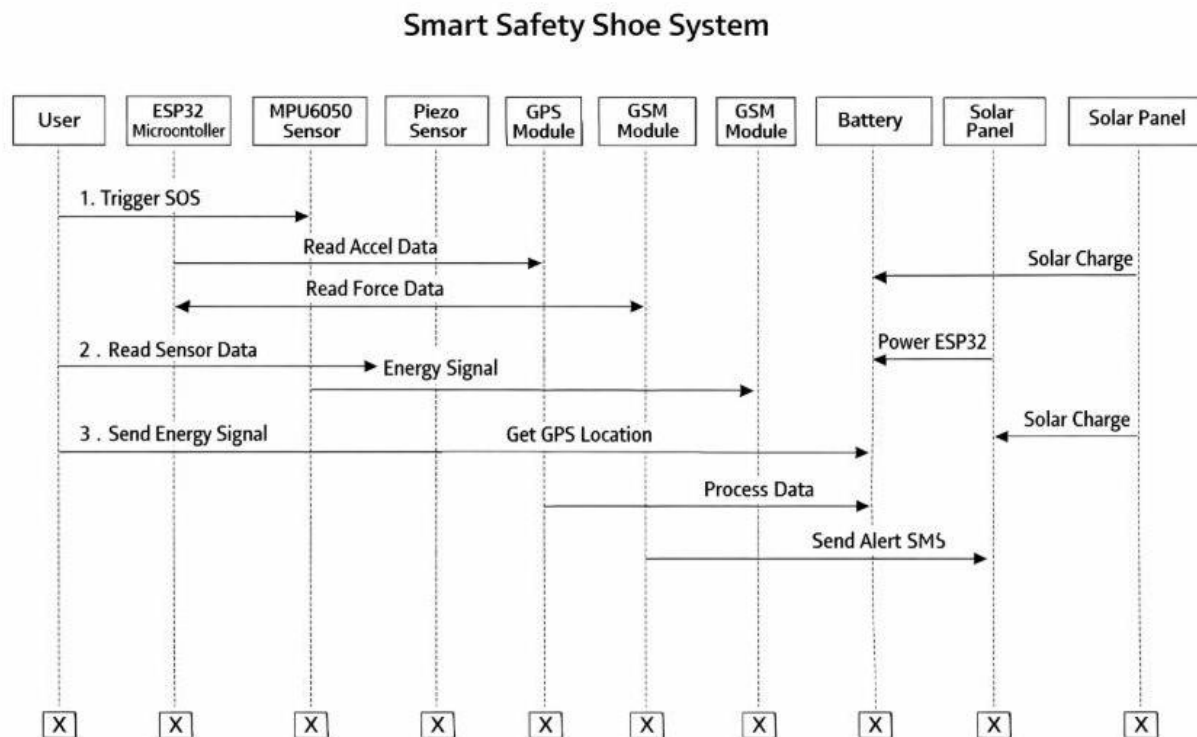
4.1.3 Sequence Diagram

A Sequence Diagram is a behavioral UML diagram that represents the interaction between different components of a system in a time sequence. It shows how messages are exchanged between various elements step-by-step to perform a particular function. This diagram helps in understanding the flow of control and communication within the system.

In the Smart Safety Shoe system, the sequence begins with the sensors continuously monitoring the user's activity. When an abnormal condition such as a fall is detected by the MPU6050 sensor, the signal is sent to the ESP32 microcontroller. The microcontroller processes the data and then requests the current location from the GPS module. After obtaining the location coordinates, the ESP32 sends this information to the GSM module, which transmits an emergency alert message to predefined contacts.

In addition to automatic detection, the sequence also includes a manual process where the user can press the SOS button. This directly triggers the microcontroller to initiate the alert sequence without waiting for sensor input. The diagram clearly shows the order of operations and interaction between components like sensors, controller, GPS, and GSM module.

Overall, the Sequence Diagram provides a clear visualization of how the system components interact over time to perform emergency detection and alerting functions.



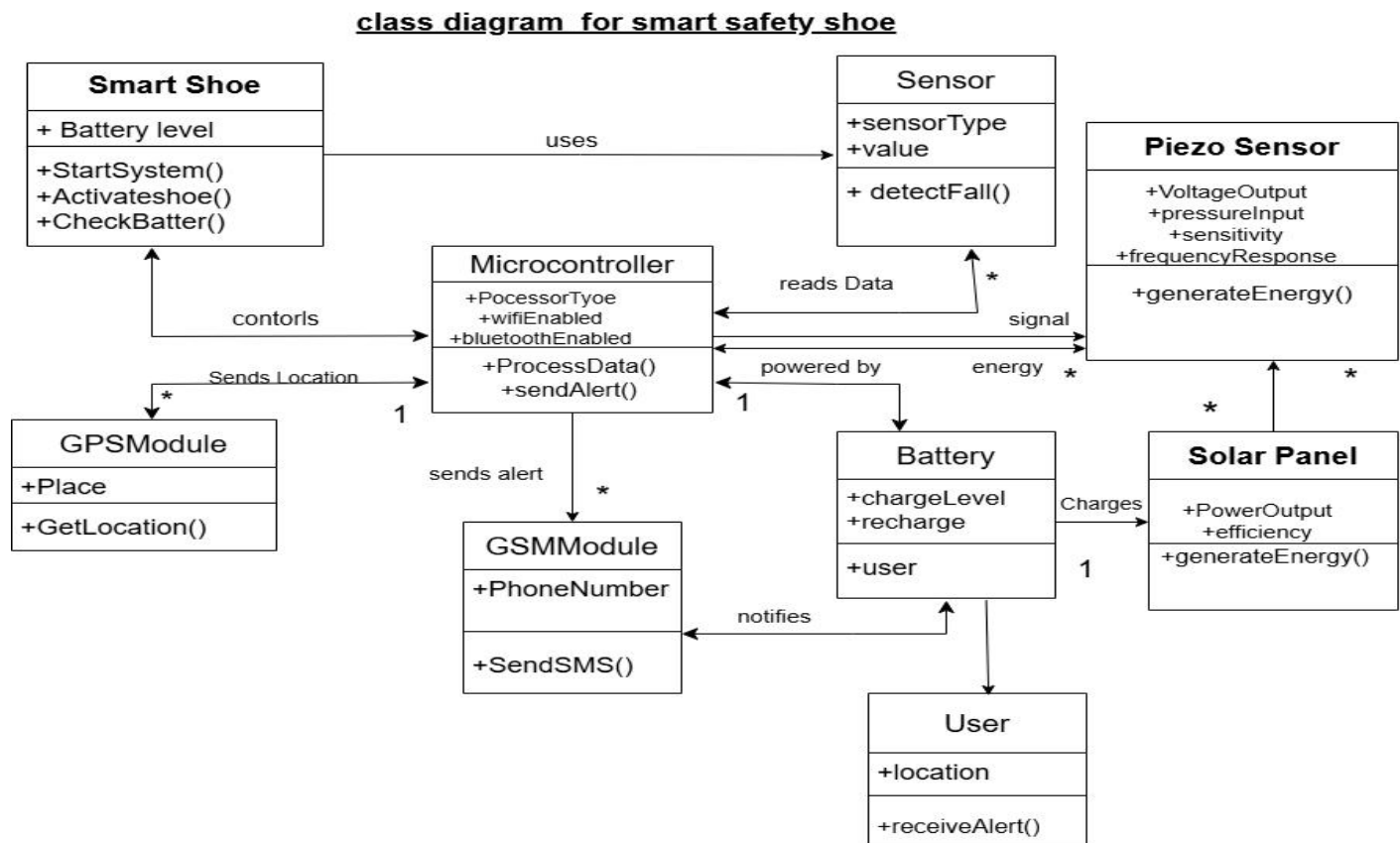
4.1.4 Class Diagram

A Class Diagram is a structural UML diagram that represents the static structure of a system. It shows the different classes involved in the system, their attributes, and the relationships between them. This diagram helps in understanding how the system is organized internally and how different components are interconnected.

In the Smart Safety Shoe system, the main classes include the Microcontroller (ESP32), sensors such as MPU6050 and piezoelectric sensor, communication modules like GPS and GSM, power components such as battery and solar panel, and output devices like LED or buzzer. Each class contains attributes that define its properties, for example, the sensor class includes parameters like sensed data, while the battery class includes charge level.

The relationships between classes are also represented in the diagram. The microcontroller acts as the central unit and is connected to multiple sensors and modules, showing a one-to-many relationship. The battery supplies power to the entire system, while energy sources like the solar panel and piezoelectric sensor are connected to the battery for charging purposes.

Overall, the Class Diagram provides a clear representation of the system architecture, showing how different components are structured and how they interact with each other at a high level.



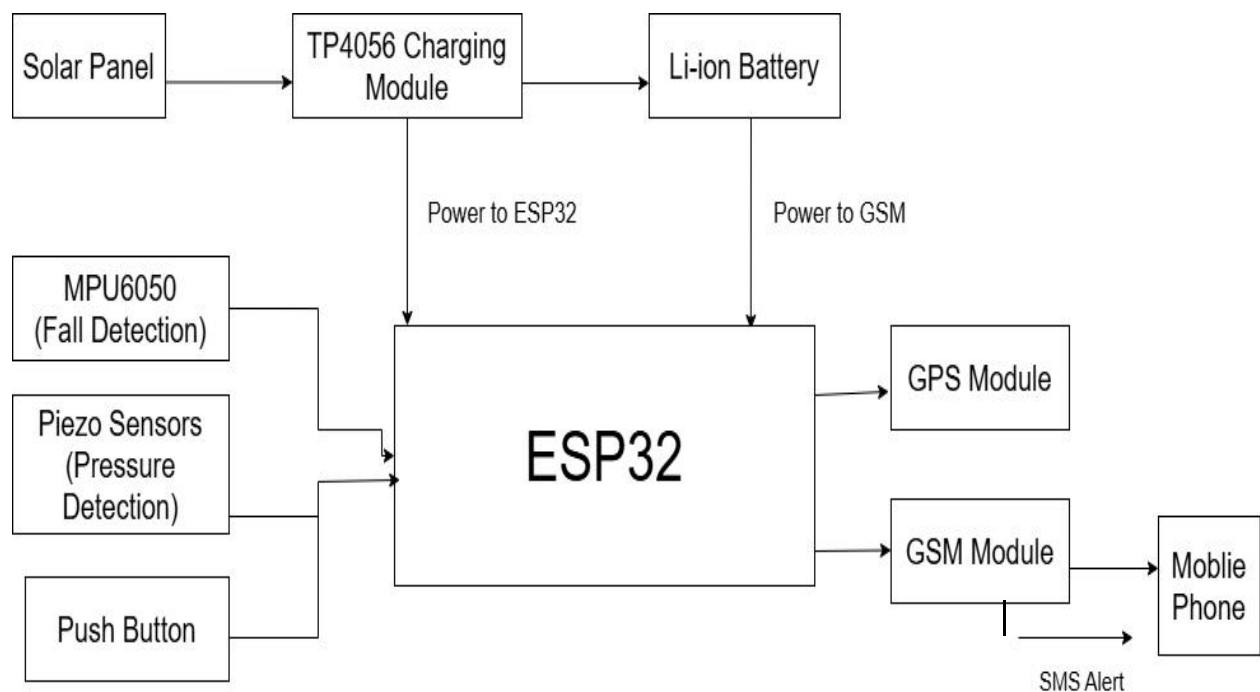
4.1.5 Component diagram

A Component Diagram is a structural UML diagram that represents the organization and interaction of different components within a system. It shows how various hardware and software modules are grouped and how they communicate with each other to perform the overall system functionality.

In the Smart Safety Shoe system, the main components include the Microcontroller (ESP32), Sensor Module (MPU6050 and Piezoelectric sensor), Communication Module (GPS and GSM), and Power Module (Battery, Solar Panel). Each component performs a specific function, such as sensing, processing, communication, or power management. The microcontroller acts as the central unit that connects all other components and controls their operations.

The Sensor Module is responsible for detecting motion and generating energy, while the Communication Module handles location tracking and sending emergency alerts. The Power Module ensures continuous power supply to the system using both battery and energy harvesting sources. The connections between these components are represented in the diagram, showing how data flows from sensors to the microcontroller and then to communication modules.

Overall, the Component Diagram provides a high-level view of the system architecture by grouping related functionalities into components, making it easier to understand the modular design and interaction of the system.



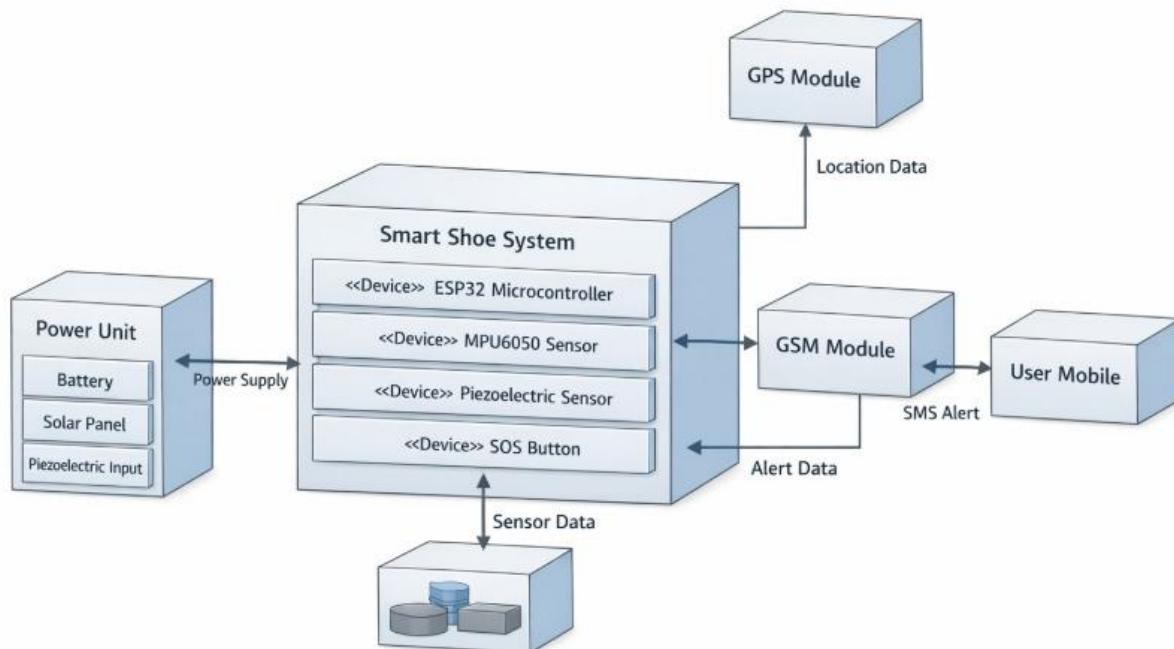
4.1.6 Deployment Diagram

A Deployment Diagram is a structural UML diagram that represents the physical arrangement of hardware components in a system and how they are connected to each other. It shows how software and hardware elements are deployed in real-world conditions and how they communicate during operation.

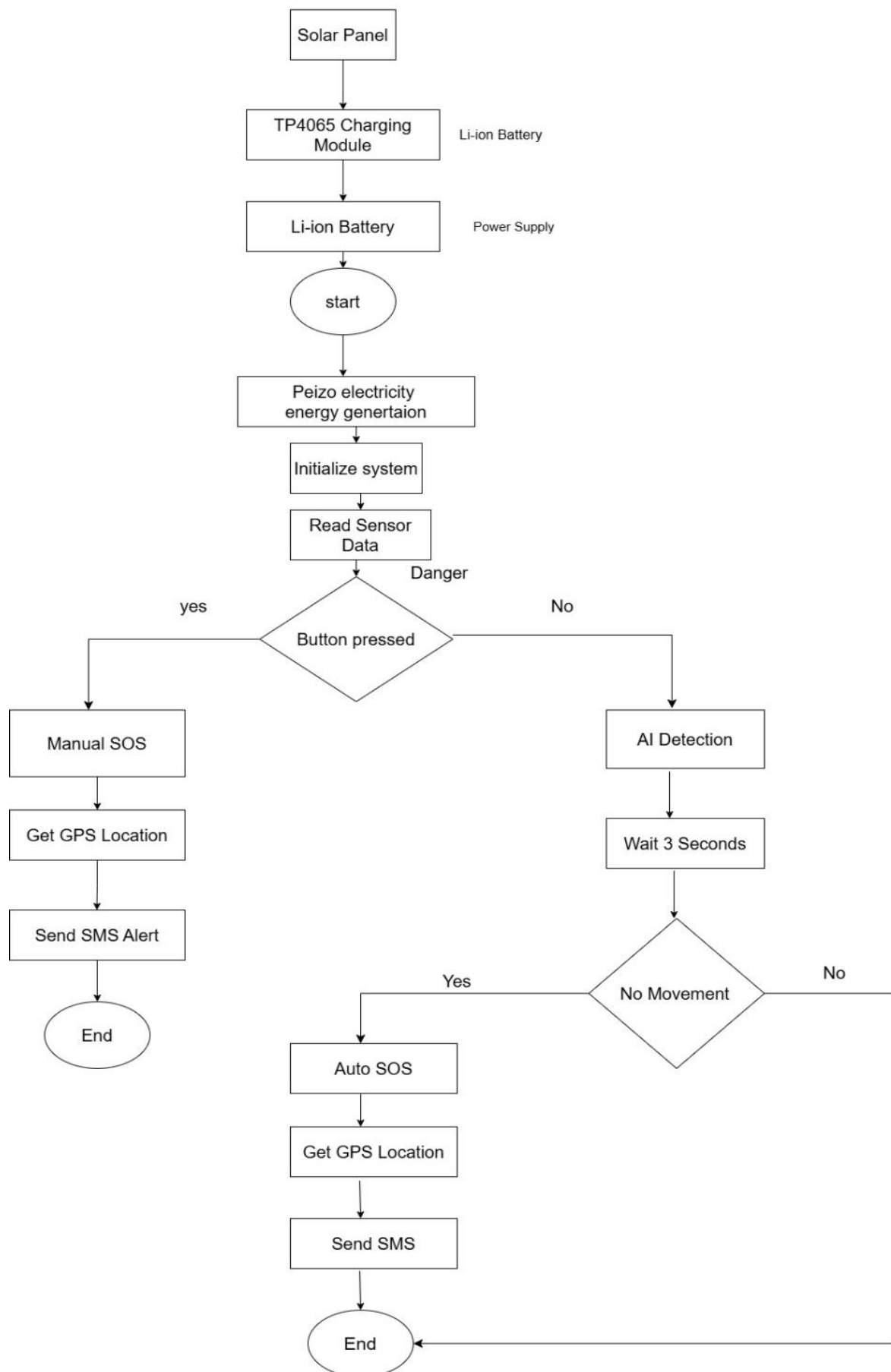
In the Smart Safety Shoe system, the deployment diagram includes components such as the ESP32 microcontroller, MPU6050 sensor, GPS module, GSM module, piezoelectric sensor, solar panel, and battery. The ESP32 acts as the central processing unit and is connected to all other modules. The sensors provide input data to the microcontroller, while the GPS and GSM modules handle location tracking and communication. The battery supplies power to the system, and it is charged using both the solar panel and piezoelectric sensor.

The diagram clearly shows the physical connections between these components, including how data flows from sensors to the microcontroller and then to communication modules. It also represents the power flow from energy sources to the battery and then to the system.

Overall, the Deployment Diagram provides a real-world view of how the system is implemented, helping to understand the hardware setup and interaction between different physical components.



4.2 FLOW CHART



4.3 SYSTEM ARCHITECTURE

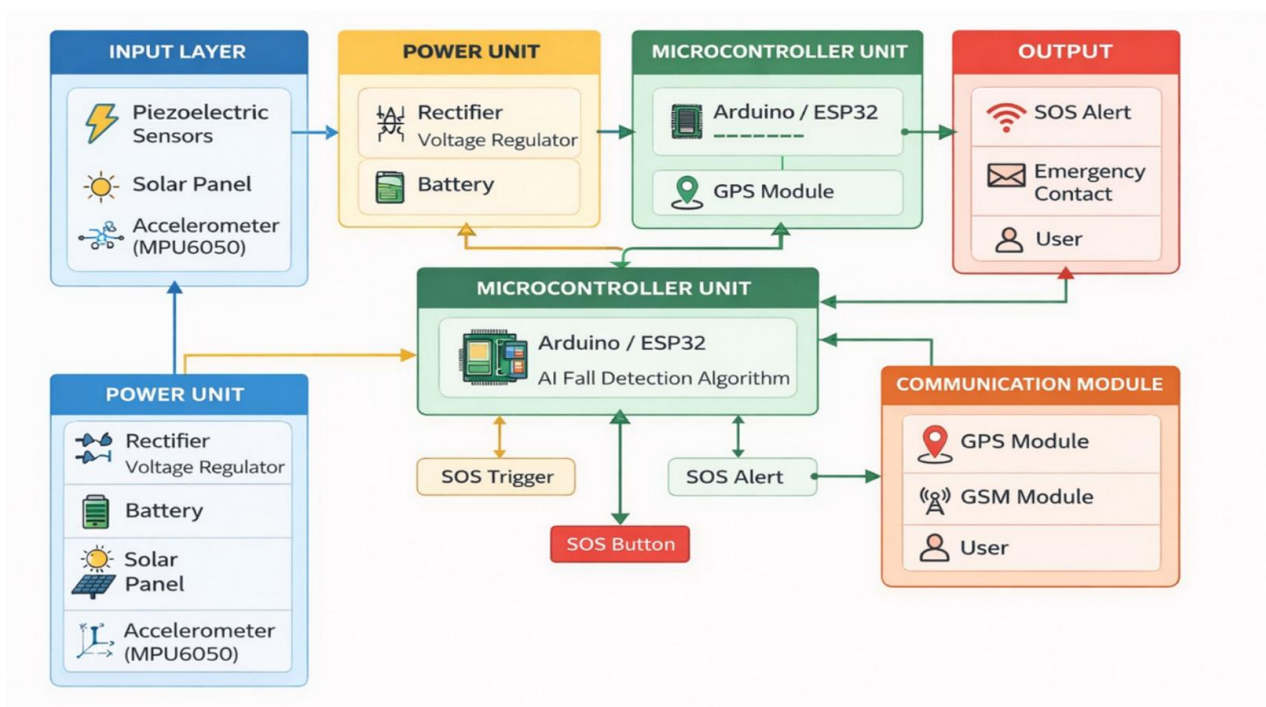
The System Architecture represents the overall structure and organization of the Smart Safety Shoe system. It provides a high-level view of how different components are interconnected and how data flows between them to achieve the desired functionality.

In this project, the ESP32 microcontroller acts as the central unit that controls and manages all operations. It is connected to various input, processing, communication, and power modules. The MPU6050 sensor continuously monitors the user's movement to detect abnormal conditions such as falls. A push button is also provided for manual emergency activation.

For communication, the GPS module is used to obtain the real-time location of the user, and the GSM module is responsible for sending alert messages to predefined contacts during emergency situations. The output devices such as LED or buzzer provide indication of system activity.

The system also includes a power management unit, where a battery supplies power to all components. The battery is charged using a hybrid energy system consisting of a solar panel and piezoelectric sensor, which generates energy from walking pressure. This ensures continuous and efficient operation of the system.

Overall, the system architecture clearly illustrates how sensing, processing, communication, and power modules are integrated into a single wearable device to provide an effective safety solution.



CHAPTER-5

IMPLEMENTATION

5.1 ARDUINO IDE

The Smart Safety Shoe system is implemented using the Arduino IDE to program the ESP32 microcontroller. Arduino IDE is a simple and user-friendly platform used for writing, compiling, and uploading code to embedded systems. It supports various libraries required for interfacing components such as MPU6050, GPS, GSM, and piezo sensors.

In this project, the Arduino IDE is used to develop the program that controls all system operations. The ESP32 reads data from sensors, processes it, and sends emergency alerts when required. The Arduino IDE plays a key role in integrating all hardware components and ensuring smooth functioning of the system.

5.2 STEPS TO INSTALL ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is required to write, compile, and upload code to the ESP32 microcontroller. The following steps describe the installation and setup process:

Step 1: Download Arduino IDE

- Visit the official Arduino website.
- Download the latest version of Arduino IDE suitable for your operating system (Windows).

Step 2: Install Arduino IDE

- Open the downloaded file.
- Follow the installation instructions.
- Click “Next” and “Install” until the setup is complete.
- Launch the Arduino IDE after installation.

Step 3: Open Preferences

- In Arduino IDE, go to:
File → Preferences
- A settings window will appear.

Step 4: Add ESP32 Board URL

- In the “Additional Board Manager URLs” field, add the ESP32 board link. This allows Arduino IDE to support ESP32 boards.
- Click OK to save.

Step 5: Install ESP32 Board

- Go to: Tools → Board → Boards Manager
- Search for “ESP32”
- Click Install for ESP32 package

Step 6: Select ESP32 Board

- Go to: Tools → Board
- Select ESP32 Dev Module

Step 7: Select Port

- Connect ESP32 to your computer using USB cable
- Go to:
Tools → Port
- Select the correct COM port

Step 8: Verify Installation

- Open a sample code (like Blink)
- Click Verify to check compilation
- If no errors, installation is successful

Step 9: Ready for Programming

- Now the Arduino IDE is ready
- You can write and upload code to ESP32

5.3 STEPS FOR UPLOADING CODE TO ESP32

Uploading code to the ESP32 microcontroller is an essential step in implementing the Smart Safety Shoe system. The following steps describe the complete process:

Step 1: Connect ESP32 to Computer

- Use a USB cable to connect the ESP32 board to your computer.
- Ensure the board is powered ON.

Step 2: Open Arduino IDE

- Launch the Arduino IDE installed earlier.
- Make sure your project code is ready.

Step 3: Write or Open Code

- Write the program code in the Arduino IDE
OR
- Open an existing project file
- Step 4: Select the Board
- Go to: Tools → Board
- Select: ESP32 Dev Module

Step 5: Select COM Port

- Go to:
Tools → Port
- Choose the correct port (e.g., COM3, COM4, etc.)

Step 6: Verify the Code

- Click on the Verify button
- This checks for errors in the code
- If errors appear, correct them before proceeding

Step 7: Upload the Code

- Click on the Upload (→) button
- The code will compile and start uploading to ESP32
- Wait until “Done Uploading” message appears

Step 8: Press Boot Button (if required)

- In some ESP32 boards, press and hold the BOOT button while uploading
- Release after upload starts

Step 9: Open Serial Monitor

- Go to:
Tools → Serial Monitor
- Set baud rate (e.g., 9600 or 115200)
- View sensor readings and output

Step 10: Execution

Once uploaded, the ESP32 starts running automatically

It begins reading sensor data and performing system functions

5.4 SOURCE CODE

The source code for the Smart Safety Shoe system is developed using Embedded C in the Arduino IDE. It controls the overall functioning of the system by integrating all hardware components such as the ESP32, MPU6050 sensor, GPS module, GSM module, piezoelectric sensor, and push button. The program begins with the inclusion of required libraries for communication and sensor interfacing. In the setup() function, all components are initialized, including serial communication, sensor modules, and input/output pins. The ESP32 establishes communication with MPU6050 using I2C protocol and with GPS and GSM modules using serial communication.

In the loop() function, the system continuously reads data from the MPU6050 sensor to monitor motion and detect abnormal conditions such as falls or no movement. It also reads analog values from the piezoelectric sensor to observe the voltage generated due to pressure.

When an emergency condition is detected or when the push button is pressed, the ESP32 retrieves the location data from the GPS module. It then sends an alert message along with location details to predefined contacts using the GSM module.

The code is designed to run continuously, ensuring real-time monitoring and quick response during emergency situations. Proper conditions and logical statements are used to control system behavior and ensure accurate functioning

```
#include <TinyGPS++.h>
#include <Wire.h>
#include <MPU6050.h>

// GPS
TinyGPSPlus gps;
HardwareSerial gpsSerial(2); // RX2 TX2 (16,17)

// GSM
HardwareSerial gsmSerial(1); // UART1 (26,27)

// MPU
MPU6050 mpu;

// Button
#define BUTTON 4

float lat = 0, lon = 0;

// AI variables
unsigned long fallTime = 0;
bool fallDetected = false;

//piezoelectric
int piezoPin = 34;
int piezoThreshold = 200; // minimum energy detection
```

```
// 📡 SEND SMS FUNCTION
void sendSMS(String message) {

    Serial.println("📡 SMS Sending...");

    gsmSerial.println("AT");
    delay(500);

    gsmSerial.println("AT+CMGF=1");
    delay(500);

    gsmSerial.println("AT+CMGS=\"+918639669611\"");
    delay(1000);

    gsmSerial.print(message);
    delay(500);

    gsmSerial.write(26); // CTRL+Z
}

void setup() {
    Serial.begin(115200);

    // GPS
    gpsSerial.begin(115200, SERIAL_8N1, 16, 17);

    // GSM
    gsmSerial.begin(9600, SERIAL_8N1, 26, 27);

    // Button
    pinMode(BUTTON, INPUT_PULLUP);

    // MPU
    Wire.begin(21, 22);
    mpu.initialize();

    Serial.println("SYSTEM READY 🚀");

    delay(10000); // allow GSM network time
}

void loop() {

    // 📍 GPS reading
    while (gpsSerial.available()) {
        | gps.encode(gpsSerial.read());
    }

    if (gps.location.isValid()) {
        | lat = gps.location.lat();
        | lon = gps.location.lng();
    }
}
```

```

// 🤖 MPU reading
int16_t ax, ay, az;
mpu.getAcceleration(&ax, &ay, &az);

float acc = abs(ax) + abs(ay) + abs(az);

// ⚪ MANUAL SOS
if (digitalRead(BUTTON) == LOW) {

    Serial.println("🚨 MANUAL SOS!");

    String msg = "🚨 EMERGENCY!\n";
    msg += "https://maps.google.com/?q=";
    msg += String(lat,6) + "," + String(lon,6);

    Serial.println(msg);
    sendSMS(msg);

    delay(5000);
}

// 🤖 FALL DETECTION
if (acc > 25000 && !fallDetected) {
    Serial.println("⚠️ FALL DETECTED!");
    fallDetected = true;
    fallTime = millis();
}

// 🤖 NO MOVEMENT CHECK
if (fallDetected && acc < 15000) {

    if (millis() - fallTime > 3000) {

        Serial.println("🚨 AUTO SOS!");

        String msg = "⚠️ EMERGENCY!\n";
        msg += "https://maps.google.com/?q=";
        msg += String(lat,6) + "," + String(lon,6);

        Serial.println(msg);
        sendSMS(msg);

        fallDetected = false;
        delay(200);
    }
}

//piezoelectricity
int piezoValue = analogRead(piezoPin);

if (piezoValue > piezoThreshold) {
    Serial.print("⚡ Energy Generated: ");
    Serial.println(piezoValue);
}
delay(200);
}

```

CHAPTER-6

RESULTS & TESTCASES

6.1 CIRCUIT DIAGRAM AND OUTPUT RESULTS

This section presents the circuit diagram and output results of the Smart Safety Shoe system. The working of each module is verified through test cases and corresponding outputs.

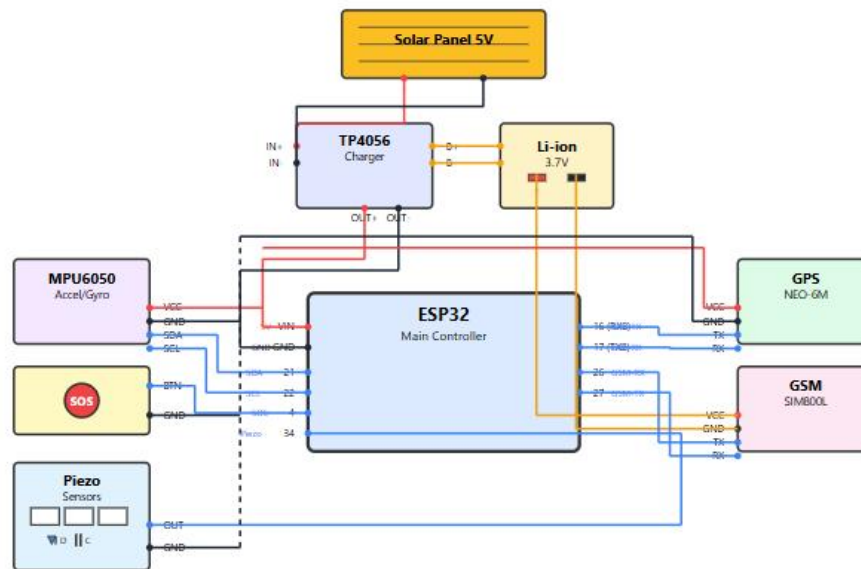


Fig 6.1 Circuit Diagram

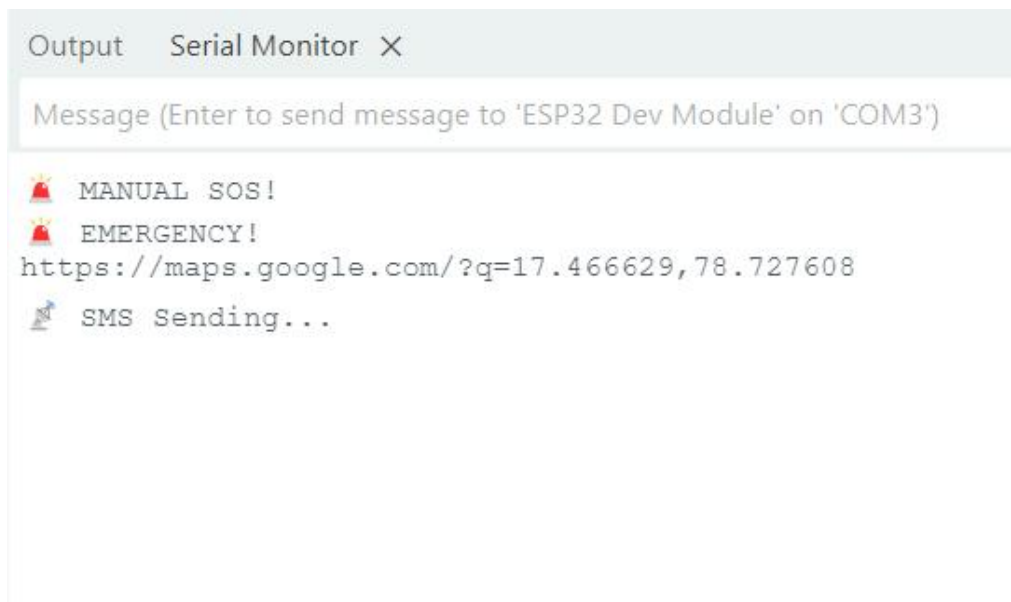


Fig 6.2 Manual SOS Trigger Output

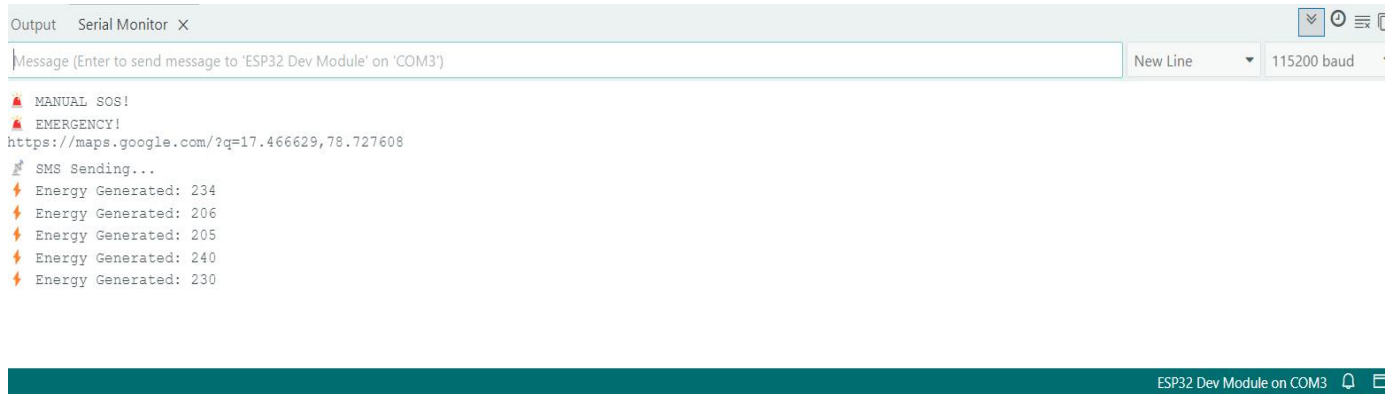


Fig 6.3 Energy Generated Output

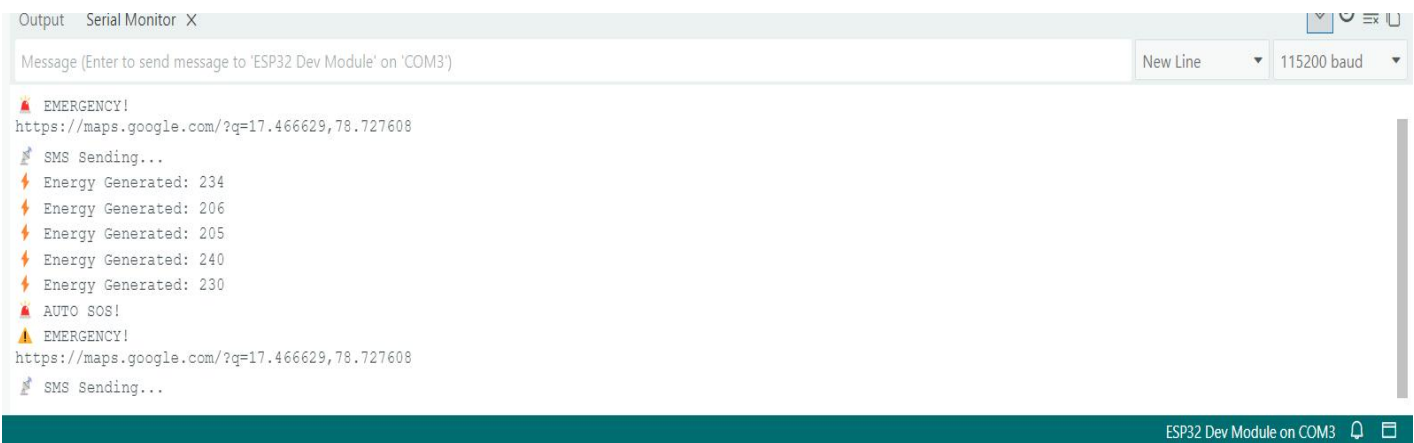


Fig 6.4 Automatic SOS Output

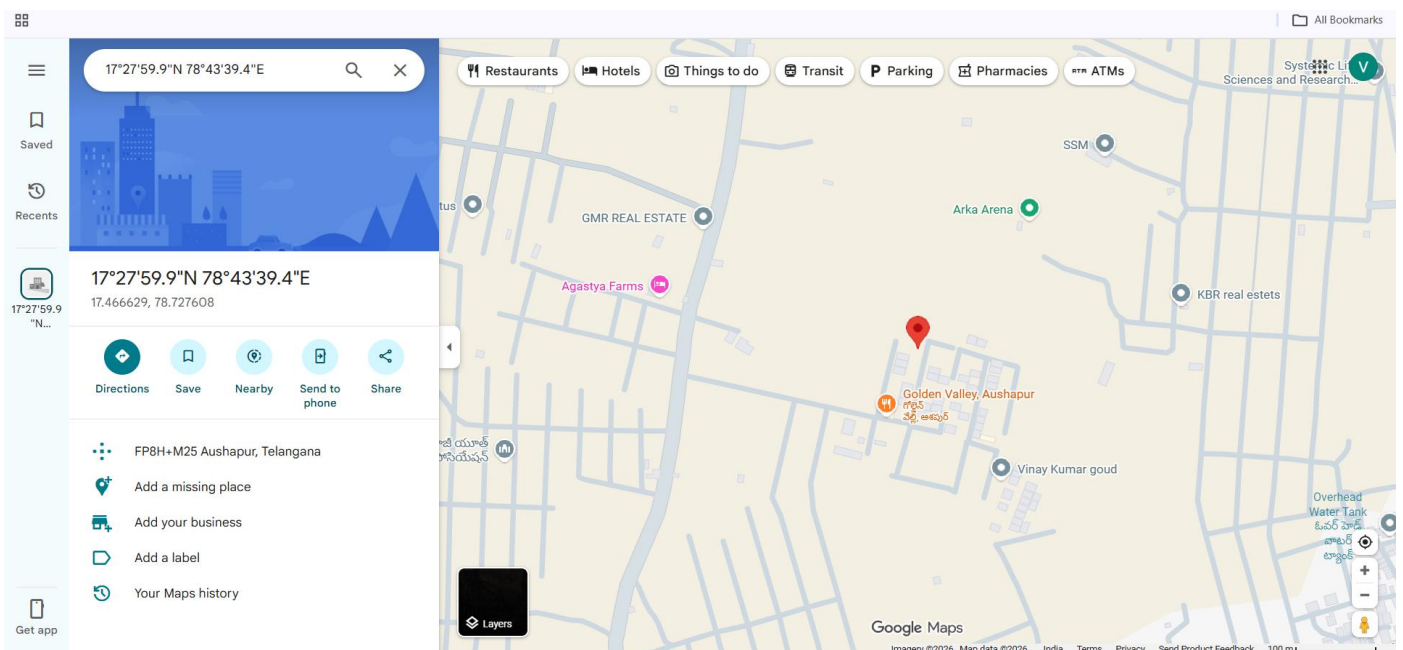


Fig 6.5 Accurate Location from GPS

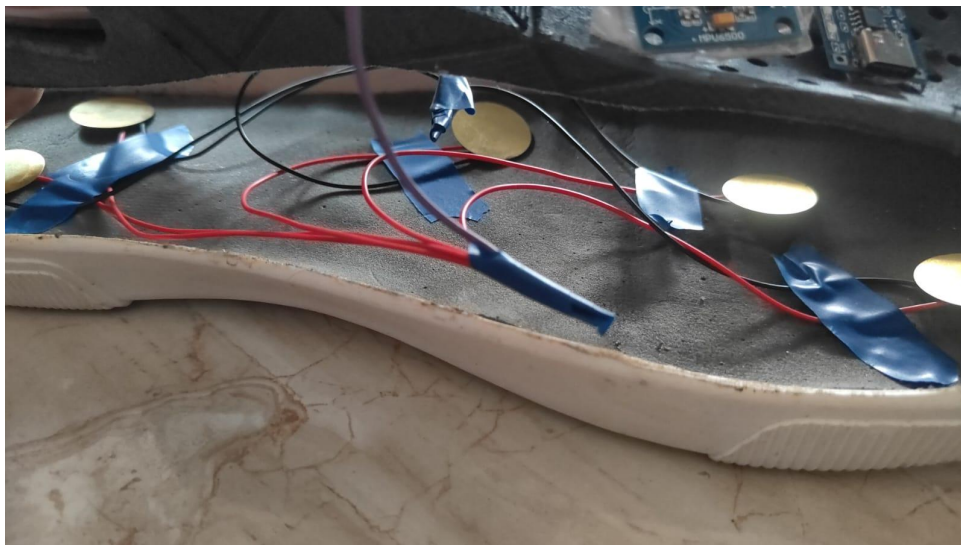
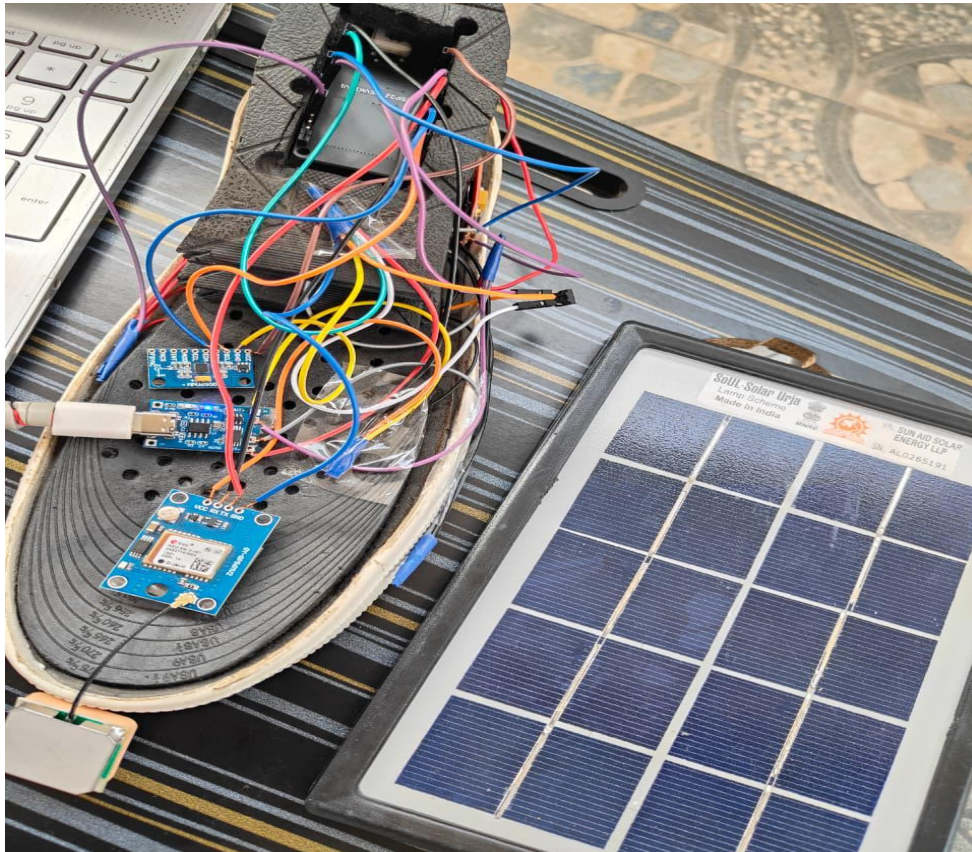


Fig 6.6 Shoe Setup with Components

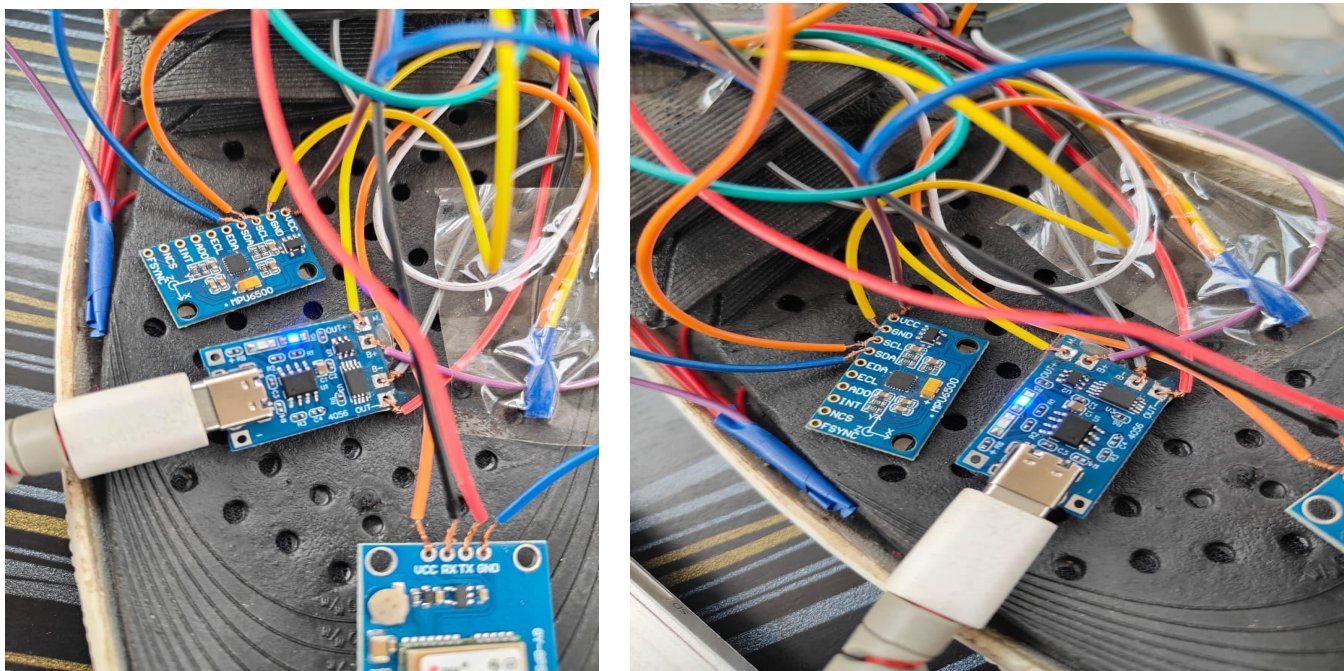


Fig 6.7 LED Indication during Operation

6.2 TEST CASE TABLE

Test Case ID	Test Case Name	Input	Process	Expected Output	Result
TC01	Fall Detection	Sudden movement / fall	MPU6050 detects abnormal motion	Emergency alert triggered	Pass
TC02	SOS Button	Button pressed	ESP32 reads button input	Alert message sent	Pass
TC03	GPS Tracking	System ON	GPS module fetches coordinates	Location displayed	Pass
TC04	GSM Alert	Emergency detected	GSM sends SMS with location	SMS received by user	Pass
TC05	Piezo Output	Walking / pressure applied	Piezo sensor generates voltage	Analog values displayed	Pass
TC06	Solar Charging	Sunlight exposure	Solar panel charges battery via TP4056	Battery level increases	Pass
TC07	System Power	Battery connected	Power supplied to ESP32 and modules	System turns ON	Pass

6.3 BENEFITS & SIGNIFICANCE

- The system significantly enhances personal safety by providing immediate emergency alerts, reducing response time during accidents or dangerous situations.
- It enables real-time monitoring of user activity using sensors, ensuring continuous observation without human intervention.
- The automatic fall detection feature improves safety by triggering alerts even when the user is unconscious or unable to act manually.
- The inclusion of a manual SOS button adds an extra layer of protection, allowing users to seek help instantly in any emergency.
- The integration of GPS and GSM technologies ensures accurate location tracking and fast communication with emergency contacts.
- The system improves response efficiency, helping rescuers quickly locate and assist the user.
- The use of hybrid energy harvesting (piezoelectric + solar) promotes sustainable energy usage and reduces dependency on traditional charging methods.
- It demonstrates the practical implementation of renewable energy in wearable electronics, making the system eco-friendly.
- The project integrates multiple domains such as embedded systems, wireless communication, and energy management, showcasing interdisciplinary innovation.
- It contributes to the development of smart wearable safety systems, which are becoming increasingly important in modern society.
- The system can help reduce risks in critical environments, improving overall user confidence and security.

6.4 ADVANTAGES

- The device is compact and lightweight, making it comfortable for daily use without affecting user mobility.
- It offers both automatic fall detection and manual SOS alert, ensuring higher reliability in emergency situations.
- Real-time GPS tracking allows quick identification of user location for faster rescue operations.
- The GSM module enables instant communication, sending alerts directly to emergency contacts.
- The use of renewable energy sources (solar + piezoelectric) improves battery efficiency and extends system operation.

- The system is cost-effective and uses easily available components, making it accessible for practical implementation.
- It can be easily upgraded and expanded with additional features like IoT and mobile applications.

6.5 DISADVANTAGES

- The energy generated by piezoelectric sensors is limited, and cannot fully power the system independently.
- The system relies on GSM network availability, which may not work effectively in remote or low-signal areas.
- Sensor accuracy may vary, leading to possible false alerts or missed detections if not properly calibrated.
- The performance of the solar panel depends on sunlight availability, reducing efficiency in indoor or low-light conditions.
- Integration of multiple components inside the shoe can be complex and space-constrained.
- Maintenance and durability of electronic components inside footwear may be a concern over long-term usage.

6.6 APPLICATIONS

- The system is highly effective for women's safety, providing immediate alerts and real-time location tracking during unsafe situations or emergencies.
- It can be widely used for elderly care, where automatic fall detection ensures timely medical assistance and reduces health risks.
- The device is suitable for individuals working in isolated or hazardous environments, such as construction workers, night shift employees, and field workers.
- It can be used by travelers and hikers, especially in unfamiliar or remote locations, ensuring safety and easy tracking.
- The system is beneficial in healthcare monitoring, where it can assist in tracking movement patterns and detecting abnormal activities.
- It can be integrated into smart city safety systems, enhancing public safety infrastructure.
- The project can be extended into fitness and activity tracking, monitoring steps, movement, and physical activity levels.
- It can be used in military or defense applications for tracking personnel in remote or risky areas.
- The system is useful for students and children safety, allowing guardians to track location and receive alerts when needed.

CHAPTER-7

DISCUSSION

7.1 OBSERVATIONS

- During the implementation of the Smart Safety Shoe system, it was observed that the integration of multiple hardware components such as ESP32, MPU6050 sensor, GPS, and GSM modules was successfully achieved, ensuring proper system functionality.
- The MPU6050 sensor effectively detected motion and was capable of identifying sudden changes such as falls. However, slight variations in readings were noticed depending on walking patterns and surface conditions.
- The GPS module was able to provide accurate location data under open sky conditions, but its performance slightly decreased in indoor environments or areas with signal obstruction.
- The GSM module successfully transmitted alert messages to predefined contacts, demonstrating reliable communication. However, message delivery speed depended on network strength.
- The SOS button functionality worked efficiently, allowing manual triggering of alerts, which adds reliability in situations where automatic detection may not be sufficient.
- The piezoelectric sensor generated electrical energy during walking, confirming the feasibility of energy harvesting. However, the output voltage varied based on pressure and intensity of movement.
- The solar panel contributed to battery charging under sunlight, improving overall energy efficiency, but its performance was limited in low-light conditions.
- The system operated effectively as a compact wearable device, though careful placement of components was required to maintain comfort and functionality.
- Overall, the system demonstrated stable performance during testing, with all modules working together to achieve the intended safety and energy objectives.

7.2 LIMITATIONS

- The energy generated by the piezoelectric sensor is relatively low and cannot independently power the entire system, making it necessary to rely on a battery as the primary power source.
- The efficiency of the solar panel is dependent on environmental conditions such as sunlight availability, which limits its performance during indoor usage or cloudy weather.
- The system depends on GSM network availability for sending emergency alerts, which may not function effectively in remote or low-signal areas.

- The accuracy of fall detection is influenced by sensor calibration and user movement patterns, which may lead to occasional false alerts or missed detections.
- Integration of multiple electronic components within the shoe requires careful design, as improper placement may affect user comfort and system durability.
- The GPS module may experience delays or reduced accuracy in enclosed environments, affecting real-time location tracking.
- Continuous operation of multiple modules can lead to higher power consumption, requiring efficient energy management.
- The system currently lacks advanced intelligence, such as AI-based decision-making, which could improve detection accuracy and adaptability.
- Maintenance and long-term durability of electronic components inside footwear can be challenging due to exposure to pressure, moisture, and environmental conditions.

CHAPTER 8

CONCLUSION

The Smart Safety Shoe project successfully presents an innovative and practical solution for enhancing personal safety using embedded systems and modern communication technologies. The system integrates multiple hardware components such as the ESP32 microcontroller, MPU6050 sensor, GPS module, GSM module, piezoelectric sensor, and solar panel into a single compact and wearable device. This integration demonstrates the effective use of interdisciplinary concepts including sensor technology, wireless communication, and energy harvesting.

The project achieves its primary objective of providing both automatic and manual emergency alert mechanisms. The automatic fall detection using the MPU6050 sensor ensures that alerts are generated even when the user is unable to respond, while the manual SOS button adds an extra layer of safety. The inclusion of real-time location tracking using GPS and instant communication through GSM significantly improves the chances of timely assistance during emergencies.

One of the key highlights of this project is the implementation of a hybrid power system. By combining piezoelectric energy generation from walking and solar energy, the system reduces dependency on conventional power sources and improves energy efficiency. This makes the system more reliable for long-term usage and aligns with sustainable energy practices.

The system is designed to be compact, lightweight, and user-friendly, making it suitable for everyday use. It is especially beneficial for women, elderly individuals, and people working in remote or isolated environments. The project also demonstrates scalability, as additional features like mobile app integration or IoT connectivity can be incorporated in the future.

In conclusion, the Smart Safety Shoe is a cost-effective, reliable, and efficient solution that successfully combines safety, communication, and energy management. The project not only meets its intended objectives but also provides a strong foundation for future advancements in wearable safety technology.

Key Achievements:

- Successfully designed and developed a Smart Safety Shoe system integrating multiple technologies into a single wearable device.
- Implemented automatic fall detection using the MPU6050 sensor, enabling emergency alerts without user intervention.
- Developed a manual SOS alert system, providing an additional safety mechanism during critical situations.
- Achieved real-time location tracking using the GPS module and successfully transmitted alerts through the GSM module.
- Integrated a hybrid energy harvesting system using piezoelectric sensors and a solar panel to improve power efficiency and sustainability.
- Ensured successful communication between all hardware components, including ESP32, sensors, GPS, and GSM modules.
- Designed a compact and wearable prototype, suitable for practical daily use.
- Completed end-to-end system implementation, including hardware setup, coding, testing, and documentation.
- Demonstrated the feasibility of combining safety, communication, and energy management in a single system.

CHAPTER -9

FUTURE SCOPE

9.1 INTRODUCTION

The Smart Safety Shoe system can be further improved by incorporating advanced technologies to enhance its performance, accuracy, and usability. Future developments can make the system more intelligent, efficient, and suitable for real-world large-scale applications.

9.2 SYSTEM ENHANCEMENTS

- Integration of a mobile application to monitor real-time data such as location, activity status, and battery level.
- Implementation of IoT connectivity for cloud storage and remote monitoring by family members or emergency services.
- Use of AI and Machine Learning algorithms to improve fall detection accuracy and reduce false alerts.
- Addition of advanced communication features such as app notifications and emergency calls for faster response.

9.3 HARDWARE AND ENERGY IMPROVEMENTS

- Miniaturization of components to make the system more compact and comfortable.
- Improvement in battery technology and energy harvesting efficiency for longer operation.
- Enhancement of solar and piezoelectric systems to move towards a self-sustaining device.

9.4 EXTENDED APPLICATIONS

- Integration of health monitoring features such as heart rate and activity tracking.
- Expansion into a multifunctional wearable device combining safety and fitness features.
- Development into a commercial product for use in safety, healthcare, and smart wearable mark

CHAPTER 10

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